

Table 39. HSE oscillator characteristics⁽¹⁾ (continued)

Symbol	Parameter	Conditions ⁽²⁾	Min	Typ	Max	Unit
$I_{DD(HSE)}$	HSE current consumption	During startup ⁽³⁾	-	-	13	mA
		$V_{DD} = 3\text{ V}$, $R_m = 30\ \Omega$, $CL = 10\text{ pF}@8\text{ MHz}$	-	0.62	-	
		$V_{DD} = 3\text{ V}$, $R_m = 45\ \Omega$, $CL = 10\text{ pF}@8\text{ MHz}$	-	0.67	-	
		$V_{DD} = 3\text{ V}$, $R_m = 30\ \Omega$, $CL = 5\text{ pF}@48\text{ MHz}$	-	1.15	-	
		$V_{DD} = 3\text{ V}$, $R_m = 30\ \Omega$, $CL = 10\text{ pF}@48\text{ MHz}$	-	1.75	-	
		$V_{DD} = 3\text{ V}$, $R_m = 30\ \Omega$, $CL = 20\text{ pF}@48\text{ MHz}$	-	5.0	-	
G_m	Maximum critical crystal transconductance	Startup	-	-	1.5	mA/V
$t_{SU(HSE)}^{(4)}$	Startup time	V_{DD} is stabilized	-	2	-	ms

1. Specified by design – Not tested in production.

2. Resonator characteristics given by the crystal/ceramic resonator manufacturer.

3. This consumption level occurs during the first 2/3 of the $t_{SU(HSE)}$ startup time

4. $t_{SU(HSE)}$ is the startup time measured from the moment it is enabled (by software) to a stabilized 8 MHz oscillation is reached. This value is measured for a standard crystal resonator and it can vary significantly with the crystal manufacturer

For C_{L1} and C_{L2} , it is recommended to use high-quality external ceramic capacitors in the 5 pF to 20 pF range (typ.), designed for high-frequency applications, and selected to match the requirements of the crystal or resonator (see [Figure 16](#)). C_{L1} and C_{L2} are usually the same size. The crystal manufacturer typically specifies a load capacitance which is the series combination of C_{L1} and C_{L2} . PCB and MCU pin capacitance must be included (10 pF can be used as a rough estimate of the combined pin and board capacitance) when sizing C_{L1} and C_{L2} .

Note: For information on selecting the crystal, refer to the application note AN2867 “Oscillator design guide for ST microcontrollers” available from the ST website www.st.com.